

IWC Migration Programme

Tranche 1 / Phase 1b

Existing-Model Corrections -- Plan vs Execution

The audit-driven correction pass of Tranche 1. Phase 1b fixes eight issues (A-H) found in a code audit of the whale-prediction ensemble, the SDM training pipeline, and the composite-risk scoring macros -- with no new external data. None of the items block the downstream density work (Phase 3), but several make it land on a sounder, honestly-calibrated base. The headline fixes: a probability NULL-deflation bug, isotonic calibration of every SDM, target-group background to cancel survey effort bias, a strike-weighted whale exposure metric, and an enforceable-protection refresh of the protection-gap sub-score.

8

Items A-H

8

SDMs recalibrated

2

Bugs fixed

2

New species

9/9

Validation PASS

270

tests pass

58.1M

Proj. rows

4

New dbt files

Status: LANDED, 2026-06-15. Phase-1b verification gate passed; Phase 2 not yet started (one chat per phase, stop at the gate).

1. Where Phase 1b Sits in the Programme

Phase 1b is the third phase of Tranche 1, after the framing work (Phase 0) and the uncertainty layer (Phase 1), and before the true vessel-traffic-density rewrite (Phase 2). Where Phase 1 ADDED an uncertainty surface, Phase 1b CORRECTS the existing point estimates -- it is a debt-paydown pass scoped entirely to models and macros that already exist, with no new external data.

Phase	Theme	Relationship to 1b
0	Framing, model cards, crosswalk	Pre-registered the 1b bugs
1	Uncertainty on existing models	Variogram feeds item E
1b	Existing-model corrections (A-H)	THIS REPORT
2	True VTD + Garrison Pleth upgrade	Rebuilds these marts next

The Phase-0 model cards deliberately pre-registered the known issues that Phase 1b now closes (uncalibrated probabilities, OBIS effort bias, the ensemble NULL-deflation bug, the fixed CV block size, the any-whale independence assumption). Phase 1b is where that backlog is paid down rather than merely documented.

Why correct now, before Phase 2 rebuilds these marts anyway

Phase 2 rebases the SAME collision-risk marts onto true VTD + the Garrison lethality curve. Doing 1b first means the rebase inherits calibrated probabilities, the NULL-aware ensemble, and the refreshed protection ladder -- so Phase 2 changes only the traffic basis, never re-litigating the whale-probability or protection science. Each phase is its own commit/verification gate per the migration plan.

2. The Plan -- Phase-1b Scope (Items A-H)

Eight audit-driven items, ordered roughly by correctness impact. Item A is a hard correctness bug; B and C are the biggest accuracy levers; D-F are validation / documentation; G and H extend the species set and the protection ladder.

Item	Title	Severity
A	Ensemble NULL-deflation bugfix	HIGH / bugfix
B	Probability calibration	HIGH
C	Target-group background + spatial thinning	MED-HIGH
D	Skill-weighted ensemble	MEDIUM
E	Variogram-driven CV block size	MEDIUM
F	any_whale independence (document)	LOW
G	Strike-weighted exposure + Rice's + gray	MED-HIGH
H	Protection-gap refresh (crit. hab + slow zones)	MEDIUM

3. The Execution -- What Was Built

A. Ensemble NULL-deflation bugfix

The ML mart ensembled four shared species as $(\text{coalesce}(\text{isd}, 0) + \text{coalesce}(\text{sd}, 0)) / 2.0$. When ISDM coverage was narrower than SDM, the missing model was treated as a hard ZERO and the sum still divided by two -- halving the

whale probability for a coverage reason, not a biological one. That deflated value then propagated into any/max/mean_whale_prob AND the Rockwood $P(\text{whale}) \times \text{traffic}$ interaction. The fix is a NULL-aware mean: average only over the models actually present (ISDM null $\rightarrow$ use SDM directly, never SDM/2). Extracted into a reusable ensemble_prob.sql macro and applied identically in the current and the projected ML marts.

B. Probability calibration

The SDMs train with $\text{scale_pos_weight} = n_neg/n_pos$, which is fine for AUC/ranking but inflates the predicted probability scale. The ML mart then MULTIPLIES $P(\text{whale}) \times \text{traffic}$ as if the values were calibrated, biasing every downstream product -- and far worse once Phase 4 uses $P(\text{whale})$ in an absolute encounter equation. The fix fits a per-species isotonic regression on out-of-fold predictions and persists it to artifacts/sdm_seasonal/<species>/calibrator.joblib. The calibrator is applied at three points so the scale stays consistent end to end: SDM training, current-grid scoring, and the CMIP6 projection scoring.

C. Target-group background + spatial thinning

The OBIS presence-background SDM already excludes traffic features, but feature exclusion alone cannot remove 'where people looked' -- effort clusters near ports, coastlines, and survey transects. Phase 1b adds target-group background (Phillips et al. 2009: draw background points from the pooled cetacean-sighting distribution so the shared effort bias cancels between presence and background) plus spatial thinning of presences (Aiello-Lammens et al. 2015) in the feature extraction and training path. This is the single biggest SDM accuracy lever in the phase.

Lower AUCs are the point, not a regression

After item C the retrained AUCs sit below the March pre-correction numbers (e.g. blue 0.684, sperm 0.628). That is expected: removing effort-bias inflation removes 'easy' separability that was really just survey geography. The new scores reflect genuine habitat signal, not where boats happened to sample.

D. Skill-weighted ensemble

The ISDM:SDM blend was a flat, unjustified 50/50 despite the two models having very different per-species reliability. Phase 1b weights the blend by each model's spatial-CV skill (Araujo & New 2007) for the four shared species, falling back to the SDM alone for the two SDM-only species (right whale, minke). The two-species SDM-only path is unchanged; the four-species shared path is now skill-weighted.

E. Variogram-driven CV block size

Spatial cross-validation previously used a fixed H3 res-2 (~158 km) block, a guess. Best practice (Valavi et al. 2019; Roberts et al. 2017) sets the block size from the residual autocorrelation range -- which is exactly the variogram that Phase 1 added. Phase 1b feeds that empirical range back into the spatial-CV block sizing, closing the loop between the two phases instead of hard-coding 158 km.

F. any_whale independence assumption (documented)

The composite $\text{any_whale_prob} = 1 - \text{product}(1 - P_i)$ assumes species independence. In reality whales co-occur (shared prey and habitat), so the formula OVERESTIMATES $P(\text{any whale})$. This is a low-severity, document-only item: the assumption is now explicitly recorded as a known simplification alongside the ensemble macro so it is not quietly forgotten when the value is consumed downstream.

G. Strike-weighted exposure + Rice's + gray whale

Two changes. First, any_whale_prob is a blunt instrument for STRIKE risk because it equal-weights a minke and a right whale. It is kept as a diagnostic, but a new strike-weighted whale exposure -- $\sum(P_i \times \text{vulnerability}_i)$ -- is promoted to feed the interaction score, with right/fin/humpback carrying high weight and minke/sperm low (strike_weighted_exposure.sql macro + strike_vuln_* vars). Second, the species set gains the two biggest omissions for a Gulf+Pacific strike platform:

- Rice's whale (Gulf of Mexico Bryde's, ~50 animals, ship strike is THE primary threat) -- wired end-to-end but DORMANT: only 8 positives, below the 50-positive training floor, so its column stays NULL and is coalesced away until survey data (GoMMAPPS) supports a model.
- Gray whale (coastal eastern-Pacific migration, regularly struck) -- fully trained and scored end-to-end (extract -> train -> score -> load -> marts), CV AUC 0.947.

A new fct_whale_vessel_exposure mart surfaces the exposure-first (raw co-occurrence) and strike-weighted views side by side, satisfying Leaper's with/without-lethality reporting rule.

H. Protection-gap refresh

The protection-gap sub-score previously tiered only MPAs + speed zones, even though critical habitat and active slow zones / DMAs had since been ingested into PostGIS. Phase 1b folds the ENFORCEABLE ones into the tier ladder via two new intermediate models (int_critical_habitat_coverage, int_slow_zone_coverage), wired into both fct_collision_risk and fct_collision_risk_seasonal with new protection_* tier vars and an updated protection_gap macro.

Why BIA stays OUT

A Biologically Important Area flags WHERE WHALES ARE, not where they are PROTECTED. Folding it into a protection score would invert the signal exactly as the proposed (non-binding) speed zones once did -- a high-whale-use area would read as 'well protected'. Only enforceable designations (critical habitat, active DMAs) enter the ladder; BIA is deliberately excluded.

4. Two Scoring Bugs Found & Fixed in Execution

Beyond the planned items A-H, materialising the projections surfaced two latent bugs in the climate-projection scoring path. Both corrupted the projected-minus-current delta -- the headline range-shift signal -- and both were fixed before the projection reload.

Bug 4 -- calibration not applied to projections

score_future_sdm.py applied the raw predict_proba (~0.27) while the current out-of-fold predictions were isotonic-calibrated (~0.019). The two scales were therefore incomparable, poisoning every projected-current delta. Fixed by loading and applying the same per-species item-B calibrator at projection-scoring time.

Bug 5 -- projections scored with stale March models

The model loader scanned a hard-coded MLflow file-store directory that held only the March pre-item-C models -- and was missing gray whale entirely. Today's item B+C runs are SQLite-backed (no file-based params), so the directory scan silently fell back to stale weights. Rewrote the loader to query the MLflow API against the SQLite store and pick the

most-recent FINISHED run per species, so the projections now use exactly the models that produced the current-grid predictions.

Result: clean, monotonic climate signal

After both fixes, projected any-whale exposure rose monotonically with warming -- SSP2-4.5: 0.0361 -> 0.0473 across decades; SSP5-8.5: 0.0378 -> 0.0631 (steeper, as the higher-emissions path should be) against a 0.0191 current baseline. The deltas are now modest and interpretable, not the broken ~0.27.

5. Design Decisions & Trade-offs

Calibrate the SAME way in all three scoring paths

The isotonic calibrator is applied at training, current-grid scoring, AND projection scoring. Using calibrated current but raw projected values (the Bug-4 state) makes the climate delta uninterpretable -- it conflates a model-scale artefact with the climate signal. Consistency of method across current and future is what makes the delta a clean attribution.

Ensemble both current AND projected the same way

The NULL-aware mean and skill weighting are applied to both the current and the projected ensembles. Mixing methods (e.g. ISDM-only current vs ISDM+SDM projected) would again corrupt the delta, so the correction is symmetric across the time axis.

Rice's whale wired-but-dormant rather than dropped

Rice's whale is the single biggest omission for a Gulf strike platform, but only 8 positive records exist -- below the 50-positive training floor. Rather than fabricate a model or omit the species, the full plumbing (crosswalk, seed, target column, mart coalesce) is in place and inert, so the day GoMMAPPS survey data lands the model drops in with no schema change.

6. Verification -- Gate Results

Both the Phase-1b exit criteria and the global gate were run. All checks passed:

Check	Command / Scope	Result
Scoring validation	validate_traffic_risk --section all	9/9 PASS
dbt build	11 table models + 93 data tests	PASS=104, 0 ERR
Projection reload	load_sdm_projections.py	58.1M rows OK
Planner stats	ANALYZE on 5 rebuilt tables	OK
Unit tests	pytest tests/	270 pass
Lint	ruff check pipeline/ backend/ tests/	clean
Orchestration	dagster definitions validate	valid
PDF	regen transformation report	16 pp OK

Gate passed

Scoring validation 9 PASS / 0 WARN; scoped dbt build PASS=104 (0 errors); 270 pytest pass; ruff clean; Dagster validates; transformation PDF regenerated. Phase 1b is complete and Phase 2 was deliberately NOT started.

7. Key Takeaways & What's Next

- Correctness before sophistication: a one-line NULL-handling bug (item A) was silently halving whale probabilities for coverage reasons -- fixing it matters more than any model tuning.
- Calibration is end-to-end or it is nothing: the value of item B was almost lost to Bug 4 because one of three scoring paths skipped it. Probability scale must be consistent everywhere it is consumed.
- Effort bias cannot be feature-engineered away: target-group background (item C) addresses 'where people looked', which excluding traffic features alone never could -- at the honest cost of lower AUCs.
- Strike risk is taxon-weighted: equal-weighting a minke and a right whale (the old any_whale view) understates the species that actually die -- item G's strike-weighted exposure fixes the signal feeding the interaction score.
- Only enforceable designations are protection: critical habitat and active DMAs enter the ladder; BIA (a presence flag) is kept out to avoid inverting the protection-gap signal.
- The MLflow file-store vs SQLite-store split is a latent trap (Bug 5) -- always load models via the MLflow API, never a directory scan.

Next: Phase 2 -- Standard-compatible traffic metric & Pleth upgrade. A true vessel-traffic-density rewrite (track-km per km², not ping counts) in `aggregate_ais.py`, a joint type x size x speed strata grain, the Garrison et al. (2025) lethality curve at two touchpoints, and a Product-A rebase of `fct_collision_risk` onto VTD -- inheriting the calibrated, NULL-aware, refreshed base that Phase 1b just established.

References

Leaper et al. (2026) SC/70/HIM/13 -- IWC strike-risk reporting standard. Miller & Kelly (2023) SC/69A/ASI/20 -- IWC model-based abundance / SDM guidance. Phillips et al. (2009) Ecol. Appl. 19:181-197 -- target-group background. Aiello-Lammens et al. (2015) Ecography 38:541-545 -- spatial thinning (spThin). Araujo & New (2007) Trends Ecol. Evol. 22:42-47 -- ensemble forecasting. Valavi et al. (2019) Methods Ecol. Evol. 10:225-232 -- blockCV. Roberts et al. (2017) -- spatial block sizing. Rockwood et al. (2021) -- co-occurrence interaction basis. Garrison et al. (2025) Front. Mar. Sci. 11:1467387 -- Pleth upgrade (Phase 2).